REMARKS ON SOBOLEV IMBEDDING INEQUALITIES [1]

B. Bojarski [2]
Department of Mathematics, University of Warsaw
PKIN, IX p., 00901 Warszawa, Poland

This paper is a continuation and extension of some topics discussed in [6]. In particular we present here a detailed proof of the theorem, only sketched in [6]. In §6 we also give another, rather direct proof of Sobolev imbedding inequalities for John domains (def. below) based on §3 of [6] and on a recent characterization of John domains given by O. Martio in [19].

We shall use the usual notations of Sobolev space theory, see e.g. [21]. For a bounded domain $\Omega \subset R^n$, $W^{l,p}_{loc}(\Omega)$ is the space of functions $f(x)$, $x \in \Omega$, with distributional derivatives $D^\alpha f$ of order $\leq l$ in $L^p(\Omega')$ for every compact subset Ω' of Ω.

The measure of a Lebesgue measurable set E is denoted by $|E|$. The average value of an integrable function f on E is denoted by the barred integral

$$\fint_E f\,dx = \frac{1}{|E|}\int_E f\,dx = f_E\,.$$

χ_E is the characteristic function of E. B and Q will be general symbols for open balls and cubes; $r(Q)$ is the sidelength of Q and σQ, $\sigma > 0$, is the cube with the same center as Q, extended homothetically σ times. P_k will denote the space of polynomials of degree $\leq k$ in n variables $(x_1, \ldots, x_n) \in R^n$.

Our starting point is the local Sobolev inequality for the functions $f(x)$ in the Sobolev space $W^{l,p}(\Omega)$, $1 \leq lp < n$, $p \geq 1$. The case $l = 1$ is of special interest because of direct geometric interpretation. Local Sobolev inequalities in this case have the form

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$$\left(\int_{R^n} |f|^{p^*} dx \right)^{1/p^*} \leq C_1(n,p) \left(\int_{R^n} |\nabla f|^p \, dx \right)^{1/p}, \quad p^* = \frac{p \cdot n}{n - p} > p \tag{1.1}$$

for f compactly supported, say $f \in C_0^\infty(R^n)$ and

$$\left(\fint_{B_R} |f - f_{B_R}|^{p^*} dx \right)^{1/p^*} \leq C_2(n,p) \, R \left(\fint_{B_R} |\nabla f|^p \, dx \right)^{1/p} \tag{1.2}$$

for functions without restrictions on the support. Here B_R is a ball with radius R and ∇f is the gradient of f. The optimal constants C_1 and C_2 in (1.1) and (1.2) depend on n and p only. (1.2) implies (1.1) with $C_1 \leq C_2$. For $p = 1$, (1.1) is known to be essentially equivalent to the isoperimetric inequality (see e.g. [21]). For general bounded domains Ω the Sobolev inequality for $l = 1$ should have the form

$$\left(\fint_{\Omega} |f - f_\Omega|^{p^*} dx \right)^{1/p^*} \leq C(n,p,\Omega) \left(\fint_{\Omega} |\nabla f|^p \, dx \right)^{1/p} \tag{1.3}$$

or the form

$$\left(\int_{\Omega} |f - f_\Omega|^{p^*} dx \right)^{1/p^*} \leq \hat{C}(n,p,\Omega) \left(\int_{\Omega} |\nabla f|^p \, dx \right)^{1/p} \tag{1.4}$$

for functions in $W^{1,p}(\Omega)$. While (1.3) and (1.4) are true for various classes of domains Ω, e.g. convex domains, domains satisfying Sobolev cone condition etc. ([30], [20], [10]) they are not true in general. Counterexamples to (1.4) go back to O. Nikodym [24], see also [21]. Inequalities (1.4) and (1.3) are closely related with the essentially weaker Poincaré inequality

$$\left(\int_{\Omega} |f - f_\Omega|^p \, dx \right)^{1/p} \leq P(n,p,\Omega) \left(\int_{\Omega} |\nabla f|^p \, dx \right)^{1/p} \tag{1.5}$$

which follows from (1.3) by the monotonicity of the arithmetic means, [13], with $P(n,p,\Omega) \leq C(n,p,\Omega)$. We remark that (1.5) is not sufficient to obtain (1.1) and is thus not equivalent to the isoperimetric inequality. We remark also, and that is in fact very essential for all what follows, that (1.3) and (1.4) are conformally invariant, in the sense that $C(n,p,\lambda\Omega) = \lambda C(n,p,\Omega)$, $\lambda > 0$ $(\hat{C}(\lambda\Omega) = \hat{C}(\Omega))$, where $\lambda\Omega$ denotes the domain obtained from Ω by homotety with dilatation λ. The Poincaré inequality (1.5) is not conformally invariant. Later we shall use (1.2) for cubes Q. The only difference is that then (1.2) holds with, in general, another constant $C_2(n,p)$. For $l > 1$ the local Sobolev inequality takes the form

$$\left(\fint_{B_R} |f - P_B(f)|^{p^*} dx \right)^{1/p^*} \leq C(n, l, p)\, R^l \left(\fint_{B_R} |\nabla^l f|^p \right)^{1/p} \tag{1.6}$$

with $p^* = \frac{np}{n - lp}$, where $\nabla^l f$ denotes the gradient of order l of f i.e. $\nabla^l f = \{D^\alpha f, |\alpha| = l\}$ and $P_B(f)$ is a polynomial of degree $\leq l - 1$. This polynomial can be chosen so that for $x \in B_R$

$$|P_B(x)| \leq C \fint_\omega |f| \tag{1.7}$$

where ω is some subset of positive measure of B_R. Then the constant C in (1.6) will depend on the ratio $|\omega|/|B_R|$. Classically $P_B(f)$ is obtained by averaging the formal Taylor expansion of f in B_R over ω. The proof of (1.6) goes back to the Sobolev papers from 1936. In the form (1.6) with the condition of type (1.7) it is contained in [10] (compare also [20], [21], [3], [4]). The local inequality (1.6) gives immediately the imbedding $W^{l,p}_{loc} \to L^{p^*}_{loc}(\Omega)$ for any domain Ω. The main result here is that for some rather broad classes of domains the local inequalities (1.6) can be "integrated" to the global inequality

$$\left(\int_\Omega |f - P_\Omega(f)|^{p^*} dx \right)^{1/p^*} \leq A \left(\int_\Omega |\nabla^l f|^p \, dx \right)^{1/p} \tag{1.8}$$

with some control, of type (1.7), for the polynomial $P_\Omega(f)$, and the coefficient A which describes the imbedding $W^{l,p}(\Omega) \to L^{p^*}(\Omega)$.

The phenomenon that in some situations the local integral estimates can be "integrated" to global estimates was probably first observed by J. Boman in [8]. He considered the case $p^* = p$, $P_\Omega(f) =$ constant. See also [4]. The novelty here is to show that the Markov type inequalities for polynomials and the scale invariance of the Sobolev inequalities make it possible to obtain a proof of a Sobolev imbedding theorem. An analogous phenomenon for Lipschitz classes was discovered by F. Gehring and O. Martio in [12]. Their result has been also applied to the Sobolev imbedding theorem for the case $p > n$: $W^{1,p}(\Omega) \to \mathrm{Lip}_\alpha(\Omega)$, see [16].

The general imbeddings $W^{l,p}(\Omega) \to C^{k,\alpha}(\Omega)$ for $lp > n$ could be also discussed via the integral estimates. This would require to consider the local (and global) inequalities of the form

$$\sum_{j=0}^{k} \|\nabla^j(f - Pf)\|_{L^q} \le C\|\nabla^l f\|_{L^p} \tag{1.9}$$

for various values of the parameters k, j, q, l, p in agreement with the general case of the Sobolev imbedding theory [21]. For this general case the full form of the Markov inequality described in §2 would be relevant, while for the case $lp < n$ and $j = 0$ only the special case $\alpha = 0$ is needed. However, even for $lp < n$, the case of admissible positive values of j, see [21], requires the Markov inequality for $|\alpha| > 0$. Details of these questions are not discussed here.

We remark also that the use of balls and cubes in our considerations is not essential. We could as well work with rectangles, triangles or even curvilinear simplexes Δ and their n-dimensional analogues with some restriction on their form so that their linear dimension is controlled by their volume in some uniform way, $|\Delta| \sim (\operatorname{diam} \Delta)^n$, and a uniform bound on the overlap after a homothetical dilatation $\Delta \to \sigma\Delta$, $\sigma > 1$. In particular for $n = 2$, "thin and long" rectangles are excluded. Generally speaking, the scale or conformal invariance of the Sobolev estimates makes it possible to use instead of cubes and balls the images of these under rather general classes of bilipschizian or even K-quasiconformal mappings, satisfying some general conditions. The stability theory of quasiconformal mappings and various global estimates of the distortion of shape and volume under these mappings, see [33], [29], [26], [25], [6], make them a good tool in studying the behaviour of various averaging processes under coordinate transformations. In particular, in [25] the distortion estimates for quasiconformal mappings have been applied to show that the BMO-class of F. John and L. Nirenberg is invariant under quasiconformal transformations.

The theory proposed here so far leaves aside the Sobolev trace theorems. There are no essential geometric reasons for that and it is possible to extend the discussed approach also to this case. The work on these problems is in progress.

2. The averaged Markov inequality

In the proof below essential use is made of the averaged Markov inequality. Various forms of Markov type inequalities can be found in the literature. The strongest and most general results in this direction seem to be the theorems of Brudnyj and Gansburg [9].

We formulate the Markov inequality in the following general form [5].

Theorem 2.1. *Let* Q *be an arbitrary cube in* R^n *and* ω *a measurable subset of* Q *of positive measure:* $|\omega| = \sigma|Q|$, $0 < \sigma \leq 1$. *Then for any polynomial* $P \in P_k$ *the inequality*

$$\left(\fint_Q |D^\alpha P|^p \, dx \right)^{1/p} \leq C|Q|^{-|\alpha|/n} \sigma^{-k} \left(\fint_\omega |P - S|^q \, dx \right)^{1/q} \tag{2.1}$$

holds, with the constant C *depending on* n, k, α, p, q *only.* S *in* (2.1) *is an arbitrary polynomial of degree strictly less than* $k = |\alpha|$. *The constants* p *and* q *are arbitrary positive numbers, the case* $p = +\infty$ *is also included.*

As shown in [5] the inequality (2.1) in its most general form can be deduced from the results of Brudnyj and Gansburg [9], who considered the case $\alpha = 0$. We remark that for our purposes it is enough to apply (2.1) to the special case when ω is a parallepiped. Then, as shown in [5], the Markov inequality (2.1) can be deduced by elementary considerations, not appealing to the results of [9]. In §4 we apply the Markov inequality for $\alpha = 0$. It then has the simpler form

$$\left(\fint_Q |P(x)|^p \, dx \right)^{1/p} \leq C\sigma^{-k} \left(\fint_\omega |P(x)|^q \, dx \right)^{1/q}, \quad \sigma = |\omega|/|Q| \tag{2.2}$$

and expresses the fact that the averages of polynomials over measurable subsets are controlled by the ratio of the measures of the corresponding sets.

Remark. Weighted averaged Markov's inequalities hold also with respect to weights satisfying the general conditions of A_p-type of Muckenhoupt [22]. See also [6].

3. Domains satisfying Boman's chain condition

Let $\sigma \geq 1$, $C > 1$ and $B > 1$ be fixed numbers (B integer).

Definition. An open set $\Omega \subset R^n$ satisfies the chain condition $F(\sigma,B,C)$ if there exist a covering F of Ω by open cubes such that

i) $\Sigma_{Q \in F} \chi_{\sigma Q}(x) \leq B\chi_\Omega(x)$, $x \in R^n$,

ii) for some fixed cube Q_0 in F, called the central cube, and for every $Q \in F$ there exists a chain $Q_0, Q_1, \ldots, Q_N = Q$ of cubes from F such that for each $\nu = 0, 1, \ldots, N-1$,

$$Q \subset BQ_\nu .$$

Moreover it is required that the consecutive cubes of the connecting chain are comparable in size and overlap in some uniform way:

iii) $$C^{-1} \leq \frac{|Q_\nu|}{|Q_{\nu+1}|} \leq C,$$

$$|Q_\nu \cap Q_{\nu+1}| \geq C^{-1} \max(|Q_\nu|, |Q_{\nu+1}|), \quad \nu = 0,1,\ldots,N-1.$$

It is important that the lengths $N = N(Q)$ of the connecting chain may depend on the particular cube in F and are not assumed to be (uniformly) bounded in the covering family F.

Many important classes of domains in R^n satisfy the chain condition. These include bounded Lipschitz domains, bounded domains satisfying the Sobolev cone condition etc. More generally, J. Boman proves in [8] the following lemma:

Lemma 3.1. *Bounded John domains satisfy the chain condition (for some choice of* B *and* C).

We recall that a domain Ω in R^n is called an (α,β)-John domain, $0 < \alpha \leq \beta$, if for some point $p_0 \in \Omega$, called a John center of Ω, each point $p \in \Omega$ can be joined to p_0 by a rectifiable (or, what is equivalent, piecewise linear) path $\gamma(t)$ of length $l(\gamma)$, $\gamma : [0, l(\gamma)] \to \Omega$, parametrized by the arc length, such that

$$l(\gamma) \leq \beta,$$

$$d(\gamma(t), \partial\Omega) \geq \frac{\alpha t}{l(\gamma)}, \quad t \in [0, l(\gamma)].$$

Here $d(p,\partial\Omega)$ is the distance of the point $p \in \Omega$ to the boundary $\partial\Omega$.

A domain Ω is a John domain if it is an (α,β)-John domain for some α and β. For an (α,β)-John domain the constants B and C depend on α, β and the dimension n. It is known that John domains may have a very irregular non-rectifiable boundary. However, important classes of domains, e.g. some irregular fractals [8], [7], are John domains. In particular, the snow-flake domains, domains bounded by a Koch curve [7], bounded quasi-discs of two-dimensional quasiconformal theory etc. are John-domains. In general domains with external cusps are not John-domains.

An important characterisation of John domains has been recently given by O. Martio [19].

4. The basic lemma

We say that the function $f(x)$, $x \in \Omega$, is in the local class $\operatorname{loc}_\sigma L^r(\Omega)$ if the integral $\int_{\sigma Q} |f|^r\,dx$ is finite for each cube $Q \subset \Omega$ such that the extended cube $\sigma Q \subset \Omega$ $(\sigma \geq 1)$. For $\sigma = 1$ we write $\operatorname{loc}_1 L^r(\Omega) = \operatorname{loc} L^r(\Omega)$.

For a real p, $1 \leq p < n/l$, $n = \dim \Omega$, $l \geq 1$, l integral, let $p^* = np/(n - lp) > p$ be the Sobolev conjugate to p.

Lemma 4.1. *Assume that the domain Ω satisfies the chain condition $F(\sigma,B,C)$. Let g be a non-negative function in the local class $\operatorname{loc}_\sigma L^p(\Omega)$. Assume that the function $f \in \operatorname{loc} L^{p^*}(\Omega)$ satisfies the following condition: for each cube $Q \in F$ there exists a polynomial $P_Q(f) \equiv P_Q$ of order at most $l-1$ such that the inequality*

$$\left(\fint_Q |f - P_Q|^{p^*} dx \right)^{1/p^*} \leq A\, r^l(Q) \left(\fint_{\sigma Q} g^p\,dx \right)^{1/p} \tag{4.1}$$

holds with a constant A not depending on the cube Q. Then there exists a constant $\hat{A}$, depending on n, p, σ, B, C, l and A only, and a polynomial $P_\Omega(f)$ of order $l-1$, such that the global inequality

$$\left(\int_\Omega |f - P_\Omega(f)|^{p^*} dx \right)^{1/p^*} \leq \hat{A} \left(\int_\Omega g^p\,dx \right)^{1/p} \tag{4.2}$$

holds if $g \in L^p(\Omega)$ i.e. $\int_\Omega g^p\,dx < \infty$.

In a rather detailed proof of the lemma we use in an essential way the ideas of the proof of Theorem 3.1 in the J. Boman's paper [8]. See also [14]. However, Sobolev type inequalities are not considered in [8].

We need also the following lemma from [8] the proof of which is given here for the sake of completeness. This lemma, as pointed in [8], is related with some ideas of O. Strömberg and A. Torchinsky in [32].

Lemma 4.2. *Let $F = \{Q_\alpha\}$ be an arbitrary family of cubes in R^n (indexed by some parameter $\alpha \in I$). Assume that for each Q_α we are given a non-negative number a_α. Then for $1 \leq p < \infty$ and $B \geq 1$ we have*

$$\left\| \sum_\alpha a_\alpha \chi_{BQ_\alpha} \right\|_{L^p} \leq D \left\| \sum_\alpha a_\alpha \chi_{Q_\alpha} \right\|_{L^p}$$

where the constant D *depends on* n, p *and* B *only.*

Proof. Let $\varphi \in L^p(R^n)$ with $1/p + 1/p' = 1$, $1 < p' \le \infty$. We will denote by $M\varphi(x) = \sup_Q \fint_Q |\varphi(t)|\,dt$ the Hardy-Littlewood maximal function of φ. Here the supremum is taken over all cubes $Q \subset R^n$ which contain x. We will use the Hardy-Littlewood inequality [31]:

$$\|M\varphi\|_{L^{p'}} \le H_{p'}\|\varphi\|_{L^{p'}} \tag{4.3}$$

with the constant $H_{p'}$ depending only on p'. Now we have, by (4.3),

$$\begin{aligned}
\Big|\int_{R^n} \sum_\alpha a_\alpha \chi_{BQ_\alpha}\varphi(x)\,dx\Big| &= \Big|\sum_\alpha a_\alpha |BQ_\alpha| \fint_{BQ_\alpha} \varphi\,dx\Big| \\
&\le B^n \sum_\alpha a_\alpha \int_{Q_\alpha} M\varphi\,dx = B^n \int_{R^n} \sum_\alpha a_\alpha \chi_{a_\alpha} M\varphi\,dx \\
&\le B^n \Big\|\sum_\alpha a_\alpha \chi_{Q_\alpha}\Big\|_{L^p} \|M\varphi\|_{L^{p'}} \\
&\le B^n H_{p'} \Big\|\sum_\alpha a_\alpha \chi_{Q_\alpha}\Big\|_{L^p} \|\varphi\|_{L^{p'}}
\end{aligned}$$

and the lemma follows with $D = B^n H_{p'}$.

Proof of Lemma 4.1. In what follows we shall use only the cubes Q belonging to the family F in the definition (3.1). We shall estimate

$$\int_\Omega |f - P_{Q_0} f|^{p^*} dx.$$

We have

$$\begin{aligned}
\int_\Omega |f - P_{Q_0} f|^{p^*} dx &\le \sum_{Q\in F} \int_Q |f - P_{Q_0} f|^{p^*} dx \\
&\le \sum_{Q\in F} 2^{p^*}\Big(\int_Q |f - P_Q f|^{p^*} dx + \int_Q |P_Q f - P_{Q_0} f|^{p^*} dx\Big) \\
&\le \tilde{A} \sum_{Q\in F} \Big(\int_{\sigma Q} g^p\,dx\Big)^{p^*/p} + 2^{p^*} \sum_{Q\in F} \int_Q |P_Q f - P_{Q_0} f|^{p^*} dx \\
&\equiv I_1 + 2^{p^*} I_2,\quad \tilde{A} = 2^{p^*} A^{p^*} \sigma^{-np^*/p}.
\end{aligned}$$

We estimate I_1 and I_2. Obviously

$$I_1 \le A\Big(\sum_{Q\in F}\int_{\sigma Q} g^p\,dx\Big)^{p^*/p} \le \tilde{A}B^{p^*/p}\Big(\int_{\Omega} g^p\,dx\Big)^{p^*/p}$$

where B is the constant from the definition (3.1).

To estimate I_2 for each pair (Q,Q_0) we use the connecting chain $\{Q_\nu\}$ $\nu = 0,1,\dots,s$ of cubes from the family F. Here $s = s(Q)$ – the length of the connecting chain – is not a priori bounded for all cubes Q in the family F. We have $Q_{s(Q)} = Q$. For each term in I_2 we have then

$$\Big(\int_Q |P_Q f - P_{Q_0}|^{p^*}dx\Big)^{1/p^*} \le \sum_{\nu=0}^{s-1}\Big(\int_Q |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx\Big)^{1/p^*} \qquad (4.4)$$

Since $Q \subset BQ_\nu$ and $Q \subset BQ_{\nu+1}$ by the condition, we have

$$\begin{aligned}\int_Q |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx &\le |BQ_\nu| \fint_{BQ_\nu} |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx \\ &\le C\,B^n|Q_\nu| \fint_{Q_\nu} |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx\end{aligned}$$

where the Markov inequality (2.2) has been applied to the polynomial $P_{Q_\nu} f - P_{Q_{\nu+1}} f$ of order $\le l-1$.

Applying the Markov inequality once more we get

$$\int_Q |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx \le B^n|Q_\nu| \int_{Q_\nu\cap Q_{\nu+1}} |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx.$$

Now we have

$$\begin{aligned}&\int_{Q_\nu\cap Q_{\nu+1}} |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*}dx \\ &\qquad \le 2^{p^*-1}\Big(\int_{Q_\nu\cap Q_{\nu+1}} |P_{Q_\nu} f - f|^{p^*}dx + \int_{Q_\nu\cap Q_{\nu+1}} |P_{Q_{\nu+1}} f - f|^{p^*}dx\Big) \\ &\qquad \le 2^{p^*-1}\Big(\int_{Q_\nu} |P_{Q_\nu} f - f|^{p^*}dx + \int_{Q_{\nu+1}} |P_{Q_{\nu+1}} f - f|^{p}\,dx\Big)\end{aligned}$$

$$\leq \tilde{A}2^{p^*-1}\Big[\Big(\int_{\sigma\tilde{Q}_\nu} g^p\,dx\Big)^{p^*/p} + \Big(\int_{\sigma\tilde{Q}_{\nu+1}} g^p\,dx\Big)^{p^*/p}\Big].$$

Finally we get

$$\Big(\int_Q |P_{Q_\nu} f - P_{Q_{\nu+1}} f|^{p^*} dx\Big)^{1/p^*} \leq G\Big[\Big(\int_{\sigma\tilde{Q}_\nu} g^p\,dx\Big)^{1/p} + \Big(\int_{\sigma\tilde{Q}_{\nu+1}} g^p\,dx\Big)^{1/p}\Big]$$

with $G = B^n C\tilde{A}2^{p^*-1}$.

The right hand side of (4.4) is then estimated by

$$2G \sum_{B\tilde{Q}\supset Q} \Big(\int_{\sigma\tilde{Q}} g^p\,dx\Big)^{1/p}$$

where the summation is extended over all cubes $\tilde{Q}$ in the family F such that $B\tilde{Q} \supset Q$. Put

$$a_Q = \frac{1}{|Q|^{1/p^*}} \Big(\int_{\sigma Q} g^p\,dx\Big)^{1/p}$$

and observe that for $x \in Q$,

$$\sum_{B\tilde{Q}\supset Q} a_{\tilde{Q}} \leq \sum_{\tilde{Q}\in F} a_{\tilde{Q}} \chi_{B\tilde{Q}}(x)$$

where in the sum on the right all cubes in F are present. Therefore the term $\int_Q |P_Q f - P_{Q_0} f|^{p^*} dx$ in I_2 is estimated by

$$\int_Q |P_Q f - P_{Q_0} f|^{p^*} dx \leq (2G)^{p^*} \Big(\sum_{B\tilde{Q}\supset Q} a_Q\Big)^{p^*} |Q| \leq (2G)^{p^*} \int_Q \Big|\sum_{\tilde{Q}\in F} a_{\tilde{Q}} \chi_{B\tilde{Q}}(x)\Big|^{p^*} dx.$$

For I_2 we get then, in view of Lemma 4.2.,

$$I_2 \leq (2G)^{p^*} \int_{R^n} \Big|\sum_{\tilde{Q}\in F} a_{\tilde{Q}} \chi_{B\tilde{Q}}(x)\Big|^{p^*} dx$$

$$\leq (2G)^{p^*} D_{p^*}^{p^*} \int_{R^n} \Big|\sum_{\tilde{Q}\in F} a_{\tilde{Q}} \chi_{\tilde{Q}}(x)\Big|^{p^*} dx$$

$$\leq (2G)^{p^*} D_{p^*}^{p^*} B^{p^*-1} \int_{R^n} \sum_{\tilde{Q}\in F} a_{\tilde{Q}}^{p^*} \chi_{\tilde{Q}}(x)\, dx$$

$$= (2G)^{p^*} D_{p^*}^{p^*} B^{p^*-1} \sum_{\tilde{Q}\in F} a_{\tilde{Q}}^{p^*} |\tilde{Q}|$$

$$= (2G)^{p^*} D_{p^*}^{p^*} B^{p^*-1} \sum_{\tilde{Q}\in F} \Big(\int_{\sigma\tilde{Q}} g^p\, dx \Big)^{p^*/p}$$

since in the sum $\Sigma_{\tilde{Q}\in F}\, a_{\tilde{Q}}\chi_{\tilde{Q}}(x)$ for each $x \in \Omega$ at most B terms are not zero. Thus

$$I_2 \leq K \Big(\sum_{\tilde{Q}\in F} \int_{\sigma\tilde{Q}} g^p\, dx \Big)^{p^*/p} \leq K B^{p^*/p} \Big(\int_{\Omega} g^p\, dx \Big)^{p^*/p}$$

with $K_1 = (2G)^{p^*} D_{p^*}^{p^*} B^{p^*-1} B^{p^*/p}$ and we get (4.2) with $\hat{A} = 2\max(A^{1/p^*}B, K^{1/p^*})$.

Remark 1. The order of the local approximating polynomials $P_Q(f)$ in the Lemma 4.1. does not need to be $l-1$. Actually the orders of the local approximating polynomials in (4.1) can be completely arbitrary (but fixed for all Q). In particular, if in the local estimates (4.1) we let the order of the polynomials be higher than $l-1$, and this is typically the case in various local approximations of solutions of partial differential equations, the finite element method etc., achieving by this some more precise information on the constants A e.g. presenting A in the form, say, $A = Dh^{\gamma}$, where h is a parameter measuring the typical linear dimension (the diameter) of the mesh of "cubes" Q (simplices Q) used in the covering F_h, and D is a factor depending on other parameters (which are dilatation invariant) describing the geometry of the covering $\{F_h\}$ as used in the proof of Lemma 4.1., then the estimate of the form $\hat{A} = Dh^{\gamma'}$ will be preserved in the global form (4.2) with some γ'.

Remark 2. A weighted version of Lemma 4.1, i.e. if on both sides of (4.1) we take the local norms in $L^{p^*}(Q,d\mu)$, and the measure $d\mu = w\, dx$ satisfies conditions of the type of A_q-conditions of Muckenhoupt [22], holds also. See [14], [23].

5. The global Sobolev inequality

The global Sobolev inequalities (1.2) and (1.6) and the Lemma 4.1 imply the general Sobolev inequality for domains satisfying the chain condition. We state the Sobolev inequality in the following form ($n > lp$):

Theorem 5.1. *Let the bounded domain* $\Omega \subset R^n$ *satisfy the chain condition. If the* L_p *norm of the* l*–th gradient* $\nabla^l f$ *of a function* $f \in W^{l,p}_{loc}(\Omega)$ *is finite then the global Sobolev inequality*

$$\left(\int_\Omega |f - P_\Omega f|^{p^*} dx \right)^{1/p^*} \le C(\Omega) \left(\int_\Omega |\nabla^l f|^p \, dx \right)^{1/p}, \quad p^* = \frac{pn}{n - lp} \tag{5.1}$$

holds with the constant $C(\Omega)$ *depending only on the parameters* C, B, σ, p *and the dimension* n *only.*

Moreover the polynomial $P_\Omega f$ (of order $\le l - 1$) can be chosen in such a way that it satisfies the inequality

$$|P_\Omega(f)(x)| \le \tilde{C}(\Omega) \int_\Omega |f| \, dx, \quad x \in \Omega \tag{5.2}$$

with the constant $\tilde{C}$ not depending on f.

Proof. If $f \in W^{l,p}_{loc}(\Omega)$ then by the local Sobolev inequality (1.6), f satisfies the assumptions of Lemma 4.1 with $g(x) = |\nabla^l f(x)|$, $\sigma = 1$. Therefore we conclude (5.1) with the polynomial $P_\Omega(f)$ equal e.g. to $P_{\Omega_0}(f)$, where Q_0 is the "central" cube of the covering F.

The choice of the polynomial $P_\Omega f$ in (5.1) allows a considerable amount of freedom. If f satisfies some additional conditions, say, if $f \equiv 0$ on an open subset ω of Ω then $P_\Omega(f)$ can be chosen $\equiv 0$. More precisely the polynomial $P_\Omega(f)$ can be always chosen in such a way that

$$|P_\Omega(f)(x)| \le K \int_\omega |f| \, dx \tag{5.3}$$

with K depending on ω (and Ω), but not on f.

Obviously, the way of choice of the polynomial $P_\Omega(f)$ may influence the optimal value of the constant $C(\Omega)$ in (5.1) so we should write $C = C(\Omega,\omega)$, but for the domains satisfying the chain condition, the estimates of the proof of Lemma 4.1 give (in general very crude, but finite) estimates for $C(\Omega)$. $P_\Omega(f)$ in (5.1) is always a projection operator on the polynomials of degree $\le l - 1$: if $f \in P_{l-1}(R^n)$ then $P_\Omega(f) \equiv f$. If f satisfies some "orthogonality conditions" then also $P_\Omega(f) \equiv 0$. The theorem 5.1 and the inequalities (5.1) and (5.2) imply the usual Sobolev inequality

$$\|f\|_{L^{p^*}(\Omega)} \le \tilde{C}\Big(\int_\Omega |f|\, dx + \Big(\int_\Omega |\nabla^l f(x)|^p\, dx \Big)^{1/p} \Big) \le \tilde{C}\|f\|_{W^{l,p}(\Omega)} \tag{5.4}$$

with the norm $\|f\|_{W^{l,p}(\Omega)}$ defined as usual.

6. The Sobolev inequality for John domains

We give here a direct proof of general Sobolev imbedding inequalities for John domains. This proof relies on some ideas discussed in [6], §3, combined with a new characterisation of John domains given by O. Martio in his interesting recent paper [18]. We first recall the precise formulation of O. Martio's result. Let A be an arbitrary subset of R^n. A mapping $f : A \to R^n$ is called an L-bilipschitz mapping if

$$|x-y|/L \le |f(x)-f(y)| \le L|x-y| \tag{6.1}$$

for all $x, y \in A$.

The open ball with center x and radius r will be denoted by $B(x,r)$, $B(0,r) = B(r)$.

Let x_0 and x be two points of the domain $\Omega \subset R^n$. We say that x lies in an (Ω,L)-bilipschitz ball $B(r)$ centered at x_0 if there exists an L-bilipschitz mapping T of $B(r)$ into Ω such that $T(0) = x_0$ and $x \in T(B(r))$. As Ω is fixed in what follows, we just use the term: L-bilipschitz ball.

The following theorem characterizes John domains.

Theorem (Martio). *If Ω is an (α,β)-John domain with center x_0, then each point $x \in \Omega$ is contained in an L-bilipschitz ball $B(\alpha)$ centered at x_0 with $L = c(\frac{\beta}{\alpha})^4$ where c is a constant depending on dimension n only. Conversely, if Ω is a domain in R^n and, for some $x_0 \in \Omega$ each point $x \in \Omega$ lies in an L-bilipschitz ball $B(\alpha)$ centered at x_0, then Ω is an $(\alpha/L^3,\alpha L)$-John domain with center x_0.*

Thus we see that John-domains Ω can be covered by a family of (Ω,L)-bilipschitzian balls $T_x(B(\alpha))$ with fixed center x_0 in Ω and fixed "radius" r, equal to the parameter α in the definition of (α,β)-John-domains. We should rather say that the L-bilipschitzian ball $T(B(r))$ has an L-radius, or quasiradius, equal to r. Notice that for any (α,β)-John domain with center x_0 the ball $B(x_0,\alpha) \subset \Omega$: moreover the ball $B(x_0,\alpha/L) = E$ is contained in the L-bilipschitzian ball $T_x(B(\alpha))$ for each $x \in \Omega$. Averages over the ball E can be used as good approximations of functions in

Sobolev classes $W^{1,l}(\Omega)$. In particular, imitating the usual Sobolev procedure to estimate the averaged oscillation $|\fint_E [f(x)-f(y)]\,dy| = |f(x)-f_E|$ by the $(n-1)$ order Riesz potentials in starshaped or convex domains, applied to the composition $f \circ T^{-1}(y)$, changing the variables in the resulting formulas, and using the estimates (6.1) the following estimate is readily obtained

$$|f(x)-f_E| \le CL^{4n} \int_A |x-y|^{1-n} |\nabla f(y)|\,dy \tag{6.2}$$

for $C \in A = T(B(R))$, i.e. for any L-bilipschitz ball.

The constant C in (6.2) depends on the dimension n only. According to the definition of $W_1(\varphi)$-domains, given in §3 of [6], the inequality (6.2) means that any L-bilipschitzian ball $A = T(B(r))$ is a $W_1(\varphi)$ domain for $\varphi = \chi_E/|E|$ equal to the normalized characteristic function of the ball E. Moreover the constant C for all these domains is uniformly bounded from above. Since obviously the union of $W_1(\varphi)$ domains, with fixed φ and fixed constant C, is also a $W_1(\varphi)$ domain, we obtain, in view of the Martio's theorem, that for any (α,β)-John domain Ω the estimate

$$|f(x)-f_E| \le C(n)L^{4n} \int_\Omega |x-y|^{1-n} |\nabla f(y)|\,dy \tag{6.3}$$

holds a.e. in Ω for all functions $f \in W^{1,1}(\Omega)$.

Thus any (α,β)-John domain is $W_1(\varphi)$-domain for $\varphi = \chi_E/|E|$ with the constant bounded from above by $L = C(n)(\frac{\beta}{\alpha})^4$. By the theorem in §3 of [6] we get then the inequality

$$|f(x)-P_{f,E}(x)| \le \hat{C} \int_\Omega \frac{|\nabla^l f|\,dy}{|x-y|^{n-l}} \quad \text{a.e. in } \Omega. \tag{6.4}$$

Thus in any (α,β)-John domain a function $f \in W^{l,1}(\Omega)$ can be approximated by a polynomial of order $\le l-1$ with the error controlled pointwise by the l-th order Riesz potentials of the l-th gradient $|\nabla^l f|$. Moreover the constant $\hat{C}$ in (6.4) depends on the dimension n, the diameter of the domain Ω, (diam Ω), and the Lipschitz constant $L = C(n)(\frac{\beta}{\alpha})^4$. This dependence is estimated from above by very explicite and simple formulas in [6]. The polynomial $P_{f,E}(x)$ is controlled by the estimate

$$|P_{f,E}(x)| \leq \hat{C}\|f\|_{W^{l-1,1}(E)} \quad \text{for } x \in \Omega \tag{6.5}$$

where $\tilde{C}$ depends on n, l and the diam Ω. In view of the fundamental Hardy-Littlewood-Sobolev theorem on fractional integration as a bounded operation in the Lebesgue L_p spaces, [2], [31], the global Sobolev imbedding inequality

$$\Big(\int_\Omega |f(x) - P_{f,E}(x)|^{p^*} dx\Big)^{1/p^*} \leq K\Big(\int_\Omega |\nabla^l f|^p \, dy\Big)^{1/p} \tag{6.6}$$

for John domains follows. The constant K in (6.6) can be estimated by the inequality

$$K \leq \hat{C}\, C(n,p,l) \tag{6.7}$$

where $C(n,p,l)$ is the L_p-norm of the Riesz $R_l(f)$ transform of order l, as an operator $R_l : L_p(R^n) \to L_{p^*}(R^n)$, $p^* = \frac{np}{n-lp}$, $n > lp$, and $\hat{C}$ is the constant from (6.4). The described procedure to obtain the Sobolev inequality (6.6) can be suitably modified to obtain estimates for the polynomials $P_{f,E}$ of the form

$$|P_{f,E}(x)| \leq \tilde{\tilde{C}} \int_E |f| \, dx.$$

with some constants $\tilde{\tilde{C}}$ not depending on f.

Also more subtle estimates of the constant $\hat{C}$ in terms of the ratio β/α are possible (with exponent $2n$ instead $4n$).

We remark that in the sketched proof of (6.6) above and the related discussion in §3 of [6], an idea of the paper [28] of J. Rešetniak is utilized. Also cf. [26], [27].

If more refined versions of Hardy-Littlewood Sobolev theorem of L_p bounds for Riesz potentials e.g. various known weighted versions of these theorems are used, [1], [22], [23], then the corresponding weighted Sobolev inequalities follow.

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